Trajectory-aware allocation of a spherical-harmonic degree budget: a non-separable formulation, an attainability benchmark, and a deployable controller

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Abstract. A spherical-harmonic evaluation costs $\mathcal{O}(N^2)$, so choosing the truncation degree along an orbit is a resource-allocation problem. Allocating from the instantaneous force defect is known to order the arcs wrongly: a trajectory error is a signed integral of the defect against the state-transition matrix, and a defect norm is a sum of magnitudes. Writing the objective in the form the dynamics impose makes the allocation problem *non-separable* in the per-epoch degrees, which is why the standard Lagrangian treatment of a budgeted separable allocation does not apply. We show that the discretized objective is a quadratic form whose coupling matrix depends on the epoch pair only through $\max(i, k)$, so a full sweep over the K_{dec} decision intervals, evaluated from M accumulation samples over $|\mathcal{N}|$ candidate degrees, costs $\mathcal{O}(M|\mathcal{N}|)$, and a Frank–Wolfe bound on the convex-hull relaxation certifies the gap; the resulting allocation is obtained without propagating anything, and every policy compared switches on one common decision grid so that the benchmark's margin is information rather than resolution. On ⟨n⟩ orbits of ⟨n⟩ independent coverage designs the certified benchmark is a median factor of ⟨value⟩ better than the equal-budget constant degree. The three criteria the paper separates are three readings of one trajectory kernel — ignore it, retain its local diagonal, retain the full cross-epoch coupling — so the question of what sensitivity weighting misses becomes the question of how much lives off the diagonal, and the answer is ⟨pct⟩. Deploying this requires the *direction* of the omitted acceleration, not only its magnitude, and that direction varies on the field's own resolution scale $\pi r/N$, so a surrogate coarser than that scale cannot represent it and one finer carries as many degrees of freedom as the omitted coefficients. We estimate it instead from the leading omitted degree bands, probing forward along the vehicle's own short-horizon prediction, and complete its amplitude from the model's measured degree-variance spectrum. That split is the paper's information statement: the spectrum is one-dimensional and costs nothing to upload, while the texture is a field at resolution $\pi r/N$ and has to be measured in flight. Covering a decision interval costs a measured ⟨pct⟩ of the gravity budget — the controller's dominant overhead, charged to it in full. The resulting controller plans a cancellation target offline on a cheap pilot arc through a forward–backward iteration and probes forward, along its own short-horizon state prediction, before committing each decision interval. Held to equal *realized* quadratic work including its own overhead, it ⟨outcome⟩ the constant degree on ⟨n⟩ of ⟨n⟩ resolved comparisons across ⟨n⟩ populations. All conclusions are model-relative and bounded by the tested designs.

Keywords: lunar gravity field; spherical harmonics; truncation degree; degree allocation; state-transition matrix; orbit propagation; GRAIL

[direction] The abstract is written to the green outcome band of `../OUTCOMES.md`. Three of its clauses invert if the campaign lands in the amber or orange band, and the alternative wordings are drafted at the head of `chapters/11_conclusion.tex` so that the paper is not rewritten under time pressure.

Nomenclature

Symbols

N	spherical-harmonic truncation degree
N_i	degree in force at accumulation epoch i
$\mathcal{N}$	candidate degree set
N_{crit}	critical-altitude fixed degree
R	reference radius of the field model
M	number of accumulation epochs
Δt_{acc}	accumulation step
Δt_{dec}	decision step (degree held constant)
$\Delta t_{\mathrm{probe}}$	tail-probe refresh interval
$\Delta\mathbf{a}$	truncation acceleration defect
$\hat{\mathbf{v}}$	estimated omitted-acceleration vector
Φ	reference state-transition matrix
$\mathbf{B}$	acceleration-to-velocity input map
$\mathbf{G}$	gravity gradient $\partial\mathbf{a}/\partial\mathbf{r}$; generates Φ
$\mathbf{u}_i$	epoch contribution $\Phi(t_0, t_i)\mathbf{B}\Delta\mathbf{a}\,\Delta t$
$\mathbf{S}_j$	partial sum $\sum_{i\le j}\mathbf{u}_i$
$\mathbf{Q}$	objective coupling matrix
$\mathbf{c}_i$	cross term $\sum_k \mathbf{Q}_{ik}\mathbf{u}_k$
λ	budget multiplier
$\mathbf{H}_r$	position selector $[\,\mathbf{I}_3\ \ \mathbf{0}\,]$
Δt_i	accumulation step at epoch i (non-uniform)
ω_j	outer-quadrature weight, $\omega_j \ge 0$, $\sum_j \omega_j = T$
$\mathbf{K}(\tau,\sigma)$, $\mathbf{K}_i$	objective kernel and its diagonal $\mathbf{K}(t_i, t_i)$; the A-SENS weight
$g(i)$, I_q	epoch-to-interval map and its fibre
W_q	interval time weight $\sum_{i\in I_q}\Delta t_i$
K_{dec}	number of decision intervals
T	arc length
k	probe depth (number of leading omitted bands)
δ	candidate-window half-width, in grid indices
Δ_{span}	degree span the window induces
n_{probe}	probe points per decision interval
r_{RHS}, N_{RHS}	right-hand-side call rate and arc total
γ	amplitude completion factor, from P_n
P_n	coefficient degree variance
J	forward–backward iteration count
L_{FW}	Frank–Wolfe lower bound
g_J, g_E	certified gap in objective and error

Budgets and metrics

B_1	nominal work rate, time average $\langle N^2\rangle_t$
B_2	realized total quadratic work $\sum_{\mathrm{RHS}} N^2$
B_3	measured serial gravity-kernel time
B_+	controller overhead
B	total nominal work, $B = B_1 T = \sum_q W_q N_q^2$
β	allocation target B_1/N_{crit}^2
β_2	realized $B_2/(K_{\mathrm{ref}} N_{\mathrm{crit}}^2)$
E	propagated seven-day position RMS error
$\hat{E}$	first-order predicted error, $\hat{E} = \sqrt{J}$
ρ, $\hat{\rho}$	error ratio, comparator over candidate
f	capture fraction of the benchmark gap
M_{res}	resolution margin; resolved if > 1
κ	probe direction accuracy $\cos\angle(\hat{\mathbf{v}}, \Delta\mathbf{a})$
κ_{eff}	the same, averaged over a decision interval
τ_{corr}	defect-direction correlation time $\pi r/(Nv)$
T_{coh}	pilot-arc coherence horizon

Policies

F-OP	operational fixed comparator
F-ENV	fixed-family lower envelope (post hoc)
R-RAD	budget-calibrated radial endpoint
R-INT	interior member
A-FORCE	force-level allocation benchmark
A-SENS	sensitivity-weighted benchmark
A-SIGN	signed trajectory-aware benchmark
C-PLAN	planned trajectory-aware controller
C-FB	budget-feedback variant
C-TGO	time-to-go-only ablation

1 Introduction

Evaluating a spherical-harmonic gravity field to degree N costs $\mathcal{O}(N^2)$, and an adaptive-step integrator makes tens of thousands of force calls per propagated day. On an eccentric orbit the required degree is not constant along the arc, so the choice of N is not one number but a schedule, and choosing that schedule under a declared compute budget is a resource-allocation problem rather than an accuracy question.

The natural way to allocate is from the local truncation defect: spend degrees where the omitted acceleration is largest. That is what an attenuation threshold, a Kaula-weighted tail criterion, or a radius-driven rule all do in different guises, and it is what a Lagrangian treatment of a budgeted separable allocation formalizes [4, 25]. It also produces the wrong ranking. Held to one gravity budget on two independent coverage designs, a schedule concentrated near perilune lowers the trajectory-averaged truncation force defect by a median factor of about five and still gives the larger seven-day position error [2, ⟨self-citation of the previous paper⟩]. The mechanism is not a sampling artefact: a first-order displacement is a *signed* integral of the defect against the state-transition matrix, and a defect norm is a sum of magnitudes, so the two agree only when the defect is incoherent, which on these arcs it is not.

[direction] The self-citation above is the previous paper. Once it has a DOI, replace the placeholder and re-check that every claim attributed to it here is one that paper actually makes at the confidence level it makes it.

That leaves an obvious question unanswered, and this paper answers it. If the force-level criterion is the wrong objective, what does the right one give, how much is it worth, and can any of it be obtained without the information an oracle has? Three questions organize the paper.

RQ1 Under trajectory-aware information, how much accuracy is *attainable* at a fixed budget, relative to the constant degree and to the best radius-driven schedules?

RQ2 How much of that comes from weighting each epoch by its sensitivity, and how much from the signed cancellation that a magnitude criterion discards?

RQ3 How much survives without the reference trajectory and without pointwise reference-field evaluations along it — the controller keeping only global, offline model metadata such as the degree variances?

Three things make the answers non-obvious, and each is a contribution.

The objective is non-separable. Writing the trajectory error in the form the dynamics impose puts the norm *after* the signed integral, which couples every pair of epochs through cancellation. The standard multiplier treatment therefore does not apply, and neither does the obvious repair of reweighting the per-epoch defect by its sensitivity: that corrects the weight while still summing magnitudes. Section 3 states the problem and Section 4 shows that it is nonetheless cheap to solve well, because the discretized coupling matrix depends on the epoch pair only through $\max(i,k)$. A full coordinate sweep costs $\mathcal{O}(M|\mathcal{N}|)$, and a Frank–Wolfe bound on the convex-hull relaxation [5, 9] certifies how far the result sits from the relaxed optimum. Nothing is propagated to obtain it. The degree is a decision variable on a coarse decision grid rather than on the fine grid the objective is accumulated over, which costs nothing in the solve and matters for the comparison: a benchmark allowed to switch faster than any deployable controller would earn part of its margin from resolution rather than from information.

Deployment needs a direction, not a magnitude. A compact tail rule predicts how much acceleration a truncation omits; it says nothing about which way it points. Cancellation is exactly the part that needs the direction, so a controller built on a tail magnitude alone can reach the sensitivity-weighted level and no further. Section 7 shows that tabulating the direction is not a cheap way out either: the omitted field varies on the model's own resolution scale $\pi r/N$, which at degree 300 is under twenty kilometres of ground track, so a surrogate coarser than that scale cannot represent it and one fine enough to do so carries a comparable number of degrees of freedom. What is affordable is to *probe* it, by evaluating the leading omitted bands — one shared stack serving every candidate degree — and to complete the amplitude from the model's measured degree-variance spectrum.

The plan and the probe have different lifetimes. The cancellation target is an accumulated quantity and is stable along an arc; the local omitted vector is field texture and is not. The same resolution scale that rules out a tabulated direction also says how far ahead one may look: over a single decision interval the vehicle's own state prediction is far more accurate than $\pi r/N$, so probing forward along it is admissible, while over a seven-day pilot arc it is not. A controller that plans offline on a cheap pilot arc and probes forward one interval at a time respects that split, reads neither the reference arc nor the field along it, and is the object the campaign of Section 8 evaluates.

Throughout, the budget that is held equal is realized total quadratic work, including the controller's own overhead, and not a nominal per-call count. The previous paper measured that the two differ by a median $\langle$ pct $\rangle$ and by a factor of $\langle$ value $\rangle$ in the compute-starved regime, which is precisely the regime where a budget matters, so the weaker definition is not used here to carry a verdict.

[direction] If the campaign lands in the amber band — benchmark gap real, controller cannot capture it — this introduction inverts: the three contributions become (i) the non-separable formulation and its certificate, (ii) the resolution-scale argument as a measured *cost* statement about surrogates for the omitted direction, and (iii) a measured account of what deployable rules leave on the table. The formulation and the decorrelation argument survive either way; only the third paragraph changes.

2 Related work

2.1 Truncation-degree selection

⟨ one paragraph ⟩ Attenuation-only thresholds, degree-variance criteria in the Meissl / Lowes–Mauersberger form [17, 24], and Kaula-type spectral rules [14]. Radius-driven selection during propagation [2]. What all of these share is that the selection functional is the local force error, evaluated pointwise.

2.2 Budgeted allocation

The separable case is classical. Allocating a scalar budget over independent items through a single multiplier is Everett's generalized Lagrange multiplier method [4], and the discrete form used for a set of quantizers is the standard treatment in rate–distortion-style bit allocation [25], which also records that over a discrete set the multiplier sweep recovers only supported efficient solutions, so a duality gap may remain. The previous paper used exactly that construction at force level.

What changes here is that the objective couples the items, so neither result applies directly. The coupled form of Eq. (8) is an integer quadratic program with a single knapsack constraint, a class with its own literature ⟨ position against the quadratic-knapsack and integer-QP literature; the citation pass supplies the references ⟩. What that literature supplies is the expectation of what is achievable — a good feasible point and a relaxation bound rather than a certified optimum — and that is what Section 4 delivers. What it does not supply is the $\max(i,k)$ structure of Eq. (9), which is specific to the objective being a quadrature over prefix sums and is what makes a sweep linear rather than quadratic in the number of epochs. From the separable literature, what carries over is the language: an *attainable* allocation reported as such rather than as a bound.

2.3 Goal-oriented error control

Weighting a local residual by the sensitivity of a functional of interest is the dual-weighted-residual idea in adaptive finite elements [3], and the distinction it draws — between a residual norm and an adjoint-weighted residual — is the same distinction the previous paper measured in this setting. Two things separate the present problem from that literature. The functional here is quadratic in an integral of the residual rather than linear in it, so the weighting is not a scalar field; and the control variable is a discrete degree with a quadratic cost, so the adaptivity decision is a budgeted integer allocation rather than a refinement flag.

2.4 Sensitivity in astrodynamics

The state-transition matrix and the variational equations that generate it are standard [26, Ch. 4]. Mapping geopotential coefficients to radial, transverse and normal position error analytically about a Kepler orbit is classical [23]. ⟨ one or two sentences ⟩ on what is different here: the mapping is used forward, as an allocation weight, and it is driven by a truncation defect rather than by an estimation residual or a single coefficient.

2.5 Multifidelity models

⟨ short paragraph ⟩ [11, 21]. The degree schedule is a fidelity schedule, and the question of where to spend fidelity under a budget is shared; what is specific here is that the low-fidelity model's error has an exploitable sign structure along the trajectory.

3 The allocation problem

3.1 Objective

Let $\mathbf{x}_{\mathrm{ref}}(t)$ be a reference trajectory on $[0,T]$ and let

$$\Delta\mathbf{a}(t,N) \;=\; \mathbf{a}_N\big(\mathbf{x}_{\mathrm{ref}}(t)\big) - \mathbf{a}_{\mathrm{ref}}\big(\mathbf{x}_{\mathrm{ref}}(t)\big) \tag{1}$$

be the truncation acceleration defect of degree N measured along it. To first order the induced *state* perturbation is $\delta\mathbf{x}(t) = \int_0^t \Phi(t,\tau)\,\mathbf{B}\,\Delta\mathbf{a}(\tau)\,\mathrm{d}\tau \in \mathbb{R}^6$, with Φ the reference state-transition matrix and $\mathbf{B}$ mapping acceleration into the velocity block [26, Ch. 4]. The metric this paper reports is a *position* RMS, so the position block must be selected before the norm is taken. With $\mathbf{H}_r = [\,\mathbf{I}_3\;\;\mathbf{0}\,]$,

$$E^2 \;=\; \frac{1}{T}\int_0^T \left\| \mathbf{H}_r \int_0^t \Phi(t,\tau)\,\mathbf{B}\,\Delta\mathbf{a}(\tau)\,\mathrm{d}\tau \right\|^2 \mathrm{d}t. \tag{2}$$

Taking the norm of the full six-vector would mix position and velocity units and would not be the quantity the campaign measures; $\mathbf{H}_r$ is carried explicitly through every expression below for that reason.

The decision variable is the degree history $N(\cdot)$, and the cost is the quadratic work $\int N(t)^2\,\mathrm{d}t$ it implies.

3.2 Why the separable treatments do not apply

Equation (2) takes the norm *after* the integral, and that is the whole of the difficulty. Once the objective is discretized, Section 3.3 puts it in the form $\mathbf{u}^\top\mathbf{Q}\,\mathbf{u}$, and the three functionals separating this paper from its predecessor are then literally three readings of *one* matrix:

$$\underbrace{\textstyle\sum_i \|\Delta\mathbf{a}_i\|^2\Delta t_i}_{\text{(i) defect norm: no }\mathbf{Q}} \qquad \underbrace{\textstyle\sum_i \Delta t_i\,\Delta\mathbf{a}_i^\top\mathbf{K}_i\,\Delta\mathbf{a}_i}_{\text{(ii) diagonal kernel}} \qquad \underbrace{\mathbf{u}^\top\mathbf{Q}\,\mathbf{u}}_{\text{(iii) all of }\mathbf{Q}} \tag{3}$$

Functionals (i) and (ii) are separable in the per-epoch degrees — (ii) because a block-diagonal quadratic form decouples — so a single multiplier decides each epoch independently [4]. Functional (iii) is not: changing N_i changes the sum against which every other epoch's contribution is measured.

Writing the ladder this way removes an arbitrary choice: the sensitivity weight in (ii) is not borrowed from anywhere, it is the objective's own coupling evaluated on its diagonal. *RQ2 is therefore exactly: how much of the gain lives off the diagonal?* The previous paper's finding — that a signed, remaining-horizon-weighted statistic still ranks two policies the wrong way round on ⟨ n ⟩ of ⟨ n ⟩ orbits — says the answer is not zero.

The diagonal restriction has to be taken in the kernel, not in the matrix. Written continuously, J is a double integral against a kernel, $J = \iint \Delta\mathbf{a}(\tau)^\top\mathbf{K}(\tau,\sigma)\Delta\mathbf{a}(\sigma)\,\mathrm{d}\tau\,\mathrm{d}\sigma$, and the set $\tau=\sigma$ has measure zero. "Dropping the off-diagonal blocks" of the discrete $\mathbf{Q}$ is therefore not a definition one can take literally: since $\mathbf{u}_i \propto \Delta t_i$, the raw diagonal term $\mathbf{u}_i^\top\mathbf{Q}_{ii}\mathbf{u}_i$ carries Δt_i^2 and shrinks as the grid is refined, so a criterion built on it would depend on the accumulation resolution — which, with the refined grid of Section 3.5, is not even uniform along the arc.

What is well defined is the kernel *on* the diagonal, a local sensitivity $\mathbf{K}(t,t)$:

$$\mathbf{K}(t,t) \;=\; \frac{1}{T}\int_t^T \mathbf{B}^\top\Phi(\tau,t)^\top\mathbf{H}_r^\top\mathbf{H}_r\,\Phi(\tau,t)\,\mathbf{B}\,\mathrm{d}\tau, \qquad \mathbf{K}_i \;=\; \frac{1}{T}\,\mathbf{B}^\top\Phi(t_0,t_i)^\top\mathbf{A}_i\,\Phi(t_0,t_i)\,\mathbf{B}, \tag{4}$$

the discrete form following from Eq. (9) and the cocycle property. Functional (ii) is then $\sum_i \Delta t_i\,\Delta\mathbf{a}_i^\top\mathbf{K}_i\Delta\mathbf{a}_i$, one power of Δt per term like (i), and it converges to a physical limit under refinement. Equivalently it is $\sum_i \mathbf{u}_i^\top\mathbf{Q}_{ii}\mathbf{u}_i/\Delta t_i$ — the raw diagonal with the extra Δt removed.

Nothing else moves. $\mathbf{K}_i$ is built from the same suffix sequence $\mathbf{A}_i$, so it costs nothing extra; and (i) and (iii) were already grid-consistent, (iii) because a double sum over $\mathbf{u}$ carries $\Delta t_i\Delta t_k$ and is a genuine double integral. The derivation of $\mathbf{Q}$, the prefix–suffix structure and the Frank–Wolfe certificate are untouched.

3.3 Discretization

Fix an accumulation grid $t_1,\ldots,t_M$ with local steps Δt_i — not assumed uniform, for the reason given in Section 3.5 — let $\mathbf{H}_r = [\,\mathbf{I}_3\;\;\mathbf{0}\,]$ select the position block, and write the transported epoch contribution and its partial sums as

$$\mathbf{u}_i(N) = \Phi(t_0,t_i)\,\mathbf{B}\,\Delta\mathbf{a}(t_i,N)\,\Delta t_i, \qquad \mathbf{S}_j = \sum_{i\le j}\mathbf{u}_i, \qquad \delta\mathbf{r}(t_j) = \mathbf{H}_r\,\Phi(t_j,t_0)\,\mathbf{S}_j, \tag{5}$$

which uses only the cocycle property $\Phi(t_j,t_i) = \Phi(t_j,t_0)\Phi(t_0,t_i)$.

Which quadrature rule the prefix sum permits. That $\mathbf{S}_j$ is a *plain* prefix sum is not a notational convenience; it requires the inner weight of epoch i to be independent of the outer epoch j, and that requirement selects the rule. A trapezoid rule for the inner integral has endpoint coefficient $(t_j - t_{j-1})/2$, which depends on j, so carrying it would replace Eq. (9) by that structure plus a j-dependent correction and forfeit the $\max(i,k)$ form. A rectangle rule keeps the structure but is first-order, which at two samples per correlation time is not a rounding detail.

We therefore place the accumulation nodes at the *midpoints* of the grid cells and sample the outer integral at the cell *edges*. The inner weight is then the full cell width — independent of j, so Eq. (9) holds verbatim — while the rule itself is the midpoint rule and second-order, and $\mathbf{S}_j$ is exactly the inner integral up to edge $j{+}1$ rather than an approximation to it. The outer integral carries no prefix sum and is free to use the trapezoid rule on the edges. Both weight sets are non-negative, which is what Section 3.4 needs. The outer integral of Eq. (2) is likewise a quadrature, with weights $\omega_j \geq 0$ summing to T, so the discretized objective is a quadratic form in the stacked vector $\mathbf{u} = (\mathbf{u}_1, \ldots, \mathbf{u}_M)$,

$$J(\mathbf{u}) \;=\; \frac{1}{T}\sum_j \omega_j\, \mathbf{S}_j^\top \mathbf{M}_j \mathbf{S}_j \;=\; \mathbf{u}^\top \mathbf{Q}\,\mathbf{u}, \qquad \mathbf{M}_j \;=\; \Phi(t_j, t_0)^\top \mathbf{H}_r^\top \mathbf{H}_r\, \Phi(t_j, t_0), \tag{6}$$

so that each $\mathbf{M}_j$ is a symmetric positive-semidefinite 6×6 block and no dimensional ambiguity is left in the notation. On a uniform grid $\omega_j = \Delta t$ and $T = M\Delta t$, and Eq. (6) reduces to the unweighted average $M^{-1}\sum_j \mathbf{S}_j^\top \mathbf{M}_j \mathbf{S}_j$; every expression below is written in the weighted form so that the two cases are one derivation rather than two.

The decision variable lives on the decision grid. The sum in Eq. (6) runs over accumulation epochs, but the degree cannot be chosen M times. Let $g(i) = q$ map an accumulation epoch to its decision interval, $I_q = \{i : g(i) = q\}$, and write

$$N_i = N_{g(i)}, \qquad W_q \;=\; \sum_{i \in I_q} \Delta t_i \tag{7}$$

for the interval's total time weight. The allocation problem is then

$$\min_{(N_1,\ldots,N_{K_{\mathrm{dec}}}) \in \mathcal{N}^{K_{\mathrm{dec}}}} \mathbf{u}^\top \mathbf{Q}\,\mathbf{u} \qquad \text{subject to} \qquad \sum_q W_q\, N_q^2 \;\leq\; B \;=\; B_1 T, \qquad K_{\mathrm{dec}} \ll M. \tag{8}$$

The budget is a time integral of the instantaneous work rate, not a count of samples: B_1 of Section 5.3 is the time average $\langle N^2 \rangle_t$, so the total over an arc of length T is $B_1 T$. Weighting by W_q rather than by the sample count $|I_q|$ is what keeps this correct when the accumulation grid is refined near perilune — an unweighted count would charge the budget as though perilune occupied most of the arc simply because it is sampled most densely. On a uniform grid $W_q = |I_q|\Delta t$ and the two agree.

This is not only an implementability constraint. Optimizing over M variables would give the benchmark more *switching freedom* than any deployable controller has, and its advantage would then be partly a statement about grid resolution rather than about trajectory-aware information. Every allocation compared in this paper — benchmark and controller alike — is piecewise constant on the same decision grid.

The result is an integer quadratic program with one knapsack constraint. Its structure, however, is unusually favourable.

3.4 The structure that makes it cheap

Because $\mathbf{S}_j$ is a prefix sum, the block of $\mathbf{Q}$ coupling epochs i and k collects exactly those j that see both, that is $j \geq \max(i, k)$:

$$\mathbf{Q}_{ik} \;=\; \frac{1}{T}\sum_{j \geq \max(i,k)} \omega_j\, \mathbf{M}_j \;=\; \frac{1}{T}\,\mathbf{A}_{\max(i,k)}, \qquad \mathbf{A}_i = \sum_{j \geq i} \omega_j\, \mathbf{M}_j. \tag{9}$$

The quadrature weights enter only inside the suffix sequence, so the structure that makes the problem tractable is untouched by a non-uniform grid. They are required to be non-negative, which rectangle and trapezoid rules satisfy and higher-order Newton–Cotes rules do not: with $\omega_j \geq 0$ and each $\mathbf{M}_j$ positive semidefinite, $\mathbf{Q}$ is positive semidefinite, and that is what the convex relaxation of Section 4.2 rests on.

$\mathbf{Q}$ therefore has no free entries at all: it is generated by the single suffix sequence $\mathbf{A}_i$, whose members are symmetric positive-semidefinite 6×6 blocks — the same size as $\mathbf{M}_j$, since $\mathbf{u}_i$ is a state perturbation and the position selection happens inside $\mathbf{M}_j$ rather than on $\mathbf{u}_i$. The cross term needed by any coordinate method follows from one prefix sum and one suffix sum,

$$\mathbf{c}_i \;:=\; \sum_k \mathbf{Q}_{ik}\mathbf{u}_k \;=\; \frac{1}{T}\Big[\, \mathbf{A}_i \sum_{k \leq i} \mathbf{u}_k \;+\; \sum_{k > i} \mathbf{A}_k\, \mathbf{u}_k \,\Big], \tag{10}$$

so evaluating $\mathbf{c}_i$ at *all* M epochs costs $\mathcal{O}(M)$ rather than the $\mathcal{O}(M^2)$ a dense $\mathbf{Q}$ would require, and the gradient $\nabla J = 2\mathbf{Q}\mathbf{u}$ is available at the same cost. A coordinate *sweep* additionally tries $|\mathcal{N}|$ candidates per decision interval and so costs $\mathcal{O}(M|\mathcal{N}|)$; the two figures are quoted separately throughout because they measure different things.

Grouping the epochs into decision intervals does not spoil this. For $I_q = \{i_1 < \cdots < i_m\}$ the within-block term telescopes, because $\max(i_p, i_{p'}) = i_p$ whenever $p' < p$:

$$\sum_{i,k\in I_q} \mathbf{u}_i^\top \mathbf{Q}_{ik}\mathbf{u}_k \;=\; \frac{1}{T}\left[\sum_p \mathbf{u}_{i_p}^\top \mathbf{A}_{i_p}\mathbf{u}_{i_p} \;+\; 2\sum_p \Big(\sum_{p'<p} \mathbf{u}_{i_{p'}}\Big)^{\!\top} \mathbf{A}_{i_p}\mathbf{u}_{i_p}\right], \tag{11}$$

which one running prefix sum evaluates in $\mathcal{O}(m)$, and the between-block term is Eq. (10) with the in-block part subtracted. A full sweep over all $K_{\rm dec}$ decision variables and all candidate degrees therefore remains $\mathcal{O}(M|\mathcal{N}|)$. Section 4 uses this.

The terminal-position objective is the special case in which the outer quadrature collapses onto the final epoch, $\omega_j = T\,\delta_{jM}$, that is

$$J_T \;=\; \big\|\,\mathbf{H}_r\,\Phi(t_M,t_0)\textstyle\sum_i \mathbf{u}_i\,\big\|^2, \tag{12}$$

with the transport matrix retained because the $\mathbf{u}_i$ have been carried back to t_0; dropping it would score the accumulated perturbation in the wrong frame. Everything below covers this case unchanged.

3.5 Two grids, not one

The decision variable and the integrand live on different time scales, and conflating them is a measurement error rather than a modelling choice.

The degree schedule $N(t)$ varies on the orbital time scale. The defect $\Delta\mathbf{a}$ does not: its direction is set by the leading omitted harmonics, whose half-wavelength at the spacecraft's radius is $\pi r/N$, so along track it decorrelates over

$$\tau_{\rm corr} \;\approx\; \frac{\pi r}{N\,v}. \tag{13}$$

A signed integral sampled coarser than that is aliased, even where the corresponding magnitude statistic is converged — and the previous paper verified convergence only of the magnitude, which moved by ⟨ pct ⟩ between a 120 s and a 10 s cadence.

Equation (13) is not one number. Both r and v swing by factors of several on the eccentric orbits studied here, and N with them, so $\tau_{\rm corr}$ runs from ⟨ value ⟩ s near a 50 km perilune at $N = 300$ to ⟨ value ⟩ s near apolune, a range of about ⟨ value ⟩. A uniform accumulation grid fine enough for perilune is therefore mostly spent where nothing happens, and one coarse enough for apolune aliases the passage that matters. The grid is refined on $\tau_{\rm corr}$ instead, which also requires the budget to be accumulated with time weights, $B_1 = \sum_i N_{g(i)}^2\,\Delta t_i/T$; on a uniform grid this reduces to the mean of Section 5.3. [WP1, WP4]

We therefore separate

- the *accumulation grid* $\Delta t_{\rm acc}$, on which $\Delta\mathbf{a}$, Φ and the partial sums are formed, chosen fine enough that J and the recovered schedule are converged (Section 5.4); and
- the *decision grid* $\Delta t_{\rm dec}$, on which N is held constant, chosen coarse enough to be implementable.

A schedule is thus piecewise constant on the decision grid while its consequences are integrated on the accumulation grid. Section 5.4 reports the convergence of both, and Section 9 reports what coarsening the decision grid costs. [WP4]

4 Solving the allocation, and certifying it

Problem (8) is solved twice: once from above, by a descent that returns an attainable schedule, and once from below, by a relaxation that bounds what any schedule could achieve. The pair is what licenses the language used for the result.

4.1 Attainable schedule: budgeted coordinate descent

Introduce the multiplier λ on the budget and minimize $J(\mathbf{u}) + \lambda\sum_q W_q N_q^2$ by sweeping the *decision intervals*. Holding all other intervals fixed, the terms involving N_q are

$$N_q \;\leftarrow\; \operatorname*{arg\,min}_{N\in\mathcal{N}} \Big[\underbrace{\sum_{i,k\in I_q} \mathbf{u}_i(N)^\top \mathbf{Q}_{ik}\,\mathbf{u}_k(N)}_{\text{within block, Eq. (11)}} \;+2\sum_{i\in I_q} \mathbf{u}_i(N)^\top \mathbf{c}_i^{(-q)} \;+\; \lambda\,W_q\,N^2\Big], \tag{14}$$

with the interval time weight W_q of Eq. (7) in place of the sample count, and $\mathbf{c}_i^{(-q)} = \mathbf{c}_i - \sum_{k\in I_q} \mathbf{Q}_{ik}\mathbf{u}_k$ from Eq. (10). The first group of terms is the block's own contribution — a magnitude, which is what a separable method would see — and the second is the coupling: it rewards a defect pointing *against* the accumulated displacement. That second term is the whole difference between this and a sensitivity-weighted allocation.

Algorithm 1 Budgeted coordinate descent for Eq. (8)

Require: tabulated $\mathbf{u}_i(N)$ for $i = 1..M$, $N \in \mathcal{N}$; suffix blocks $\mathbf{A}_i$; grouping g; budget B; start set $\mathcal{S}$ (Table 1)

$\lambda \leftarrow$ bracket from the separable solution
repeat ▷ outer: bisection on the budget
  for each start $s \in \mathcal{S}$ **do** ▷ multi-start, *not* one start
    $N \leftarrow s$
    repeat ▷ inner: sweeps
      refresh prefix sums of $\mathbf{u}$ and suffix sums of $\mathbf{A}_k\mathbf{u}_k$
      for $q = 1 \ldots K_{\text{dec}}$ **do**
        update N_q by Eq. (14); patch the two running sums over I_q
      end for
    until no interval changes, or sweep limit
    record $(J(N), N)$
  end for
  $N^\star(\lambda) \leftarrow \arg\min_s J$ ▷ best of starts, at this λ
  $\lambda \leftarrow$ bisect on $\sum_q W_q N_q^{\star 2} - B$
until budget met to tolerance ⟨value⟩
return $N^\star$, $J(N^\star)$, and the spread of J across $\mathcal{S}$

The multi-start loop sits *inside* the bisection rather than outside it: each λ is solved from scratch from all starts. That is more expensive than warm-starting from the previous λ, and it is deliberate — a warm start would make the returned schedule depend on the bisection path, which is precisely the failure mode the monotonicity check below is meant to detect.

Each sweep is $\mathcal{O}(M|\mathcal{N}|)$ by Eqs. (10) and (11) — grouping changes the number of decision variables but not the sweep cost — and no field evaluation occurs inside the loop: the $\mathbf{u}_i(N)$ table is built once per orbit.

Summation. The prefix sums are accumulated in compensated (Neumaier) arithmetic. This is not routine caution: $\mathbf{S}_j$ is a sum of order 10^5 signed terms whose cancellation is the very quantity the method exploits, so the summation is ill-conditioned by construction, and the ratio of $\sum_i \|\mathbf{u}_i\|$ to $\|\mathbf{S}_M\|$ is reported per orbit as the conditioning diagnostic. [WP1]

The objective is not convex in the integer variables, so the descent is started from several points and the spread across starts is reported rather than suppressed [WP3].

The outer bisection needs a monotonicity check. For a global minimizer the attained work $W(\lambda)$ is non-increasing in λ, which is what licenses bisecting on it. What Algorithm 1 returns is a best-of-starts local solution, and nothing guarantees that the basin it lands in varies monotonically. We therefore sweep λ on a dense logarithmic grid before any campaign run and verify that the best-of-starts $W(\lambda)$ is non-increasing. Where it is, bisection is used. Where it is not, the bisection is abandoned for that orbit and the schedule is chosen as the feasible grid point with the smallest J, which is a weaker procedure but not an unsound one; the orbits on which this happens are counted and reported. [WP3]

Table 1. Starting schedules for Algorithm 1 and the spread of the attained objective across them. Declared before the runs.

Start	Rationale	Median J/J_{best}
F-OP	constant degree at the budget	⟨value⟩
R-RAD	budget-calibrated radial endpoint	⟨value⟩
R-INT	interior member	⟨value⟩
A-SENS	separable sensitivity-weighted solution	⟨value⟩
random $\times 8$	spread	⟨value⟩–⟨value⟩

4.2 Certificate: convex-hull relaxation and a Frank–Wolfe gap

Calling the descent result a benchmark requires knowing how far it can be from the best possible schedule. Relax each *decision interval* to the convex hull of its candidate contributions,

$$\mathbf{u}_i \;=\; \sum_{N\in\mathcal{N}} \theta_{g(i)N}\,\mathbf{u}_i(N), \qquad \theta_{qN}\ge 0, \quad \sum_N \theta_{qN}=1, \qquad \sum_q W_q \sum_N \theta_{qN}N^2 \;\le\; B. \tag{15}$$

Every integer schedule is feasible here with θ an indicator and attains the same objective, so the relaxed optimum J^*_{rel} is a lower bound on the integer optimum J^*. The relaxed problem minimizes the convex quadratic J over a convex set, and its linear subproblem

$$\mathbf{s} \;\in\; \operatorname*{arg\,min}_{\theta\ \text{feasible}} \langle \nabla J(\mathbf{u}), \mathbf{u}(\theta)\rangle \tag{16}$$

decouples across decision intervals under a single multiplier on the budget. It therefore has the same epoch-separable, single-multiplier *structure* as the sensitivity-weighted allocation of Section 6.1, but it is not the same problem: its per-interval objective is $\langle \nabla_q J, \mathbf{u}_q(N)\rangle$ rather than $\sum_{i\in I_q} \Delta t_i \Delta\mathbf{a}_i^\top \mathbf{K}_i \Delta\mathbf{a}_i$, and $\nabla J = 2\mathbf{Q}\mathbf{u}$ depends on the current relaxed iterate. Frank–Wolfe therefore applies with an $\mathcal{O}(M|\mathcal{N}|)$ iteration [5, 9], and its duality gap supplies the bound directly:

$$J(\mathbf{u}) + \langle \nabla J(\mathbf{u}), \mathbf{s}-\mathbf{u}\rangle \;\le\; J^*_{\mathrm{rel}} \;\le\; J^* \;\le\; J(\mathbf{u}_{\mathrm{descent}}). \tag{17}$$

Write L_{FW} for the best (largest) value the left-hand side attains over the iteration budget and J_{desc} for the descent objective. Because the objective is a squared error, a gap in J and a gap in E are not the same number, so both are defined and the threshold is fixed on the second:

$$g_J \;=\; \frac{J_{\mathrm{desc}} - L_{\mathrm{FW}}}{J_{\mathrm{desc}}}, \qquad g_E \;=\; 1-\sqrt{\frac{L_{\mathrm{FW}}}{J_{\mathrm{desc}}}} \;=\; 1-\sqrt{1-g_J}. \tag{18}$$

g_E is the quantity the paper's claims are in — it states that the descent schedule's position error is within g_E of the best error any schedule of this class could achieve — and the naming rule below is bound to it. A 10% error gap corresponds to $g_J \approx 19\%$.

Two degenerate cases are handled by rule rather than by judgement. The Frank–Wolfe bound is valid but can be vacuous early in the iteration, where the left-hand side of Eq. (17) may fall below zero; since $J \ge 0$ always, L_{FW} is taken as $\max\{0,\ \text{best bound}\}$. If $L_{\mathrm{FW}} = 0$ when the iteration budget is exhausted, the certificate is vacuous, no gap is reported for that orbit, and the orbit is counted in a separate column rather than dropped.

The subproblem must be solved exactly. The bound in Eq. (17) is valid only if $\mathbf{s}$ attains the minimum of Eq. (16) over the *relaxed* feasible set. Stopping the multiplier bisection at the nearest schedule whose integer budget falls below B does not do that: it returns a feasible point, not a minimizer, and the resulting quantity is no longer a lower bound. Because Eq. (15) permits fractional mixing, the budget can be met with equality — at the critical multiplier the solution mixes the two adjacent degrees of the affected intervals with the weights that saturate the constraint. The implementation constructs that mixture explicitly and asserts $|\sum_q W_q \sum_N \theta_{qN}N^2 - B|$ at machine precision on every iteration; a run in which the assertion fails produces no certificate.

What the certificate does and does not license. It bounds the distance to the best schedule *of this problem*: the first-order displacement along a fixed reference trajectory, on a fixed accumulation grid, over a discrete degree set. It says nothing about the nonlinear trajectory, which Section 8.8 tests separately. We adopt the naming rule fixed before the runs: the quantity is called an *oracle* only if the median g_E of Eq. (18) is below 0.10 and no more than ⟨ n ⟩ orbits return a vacuous certificate; otherwise it is reported as a *linearized trajectory-aware allocation benchmark* and every ratio taken against it is read as conservative, in the same sense the previous paper's force-level Lagrangian benchmark was.

[direction] One of the two names has to be chosen once Table 2 exists, and the choice is bound to the gap number rather than to the result. If the gap is large, delete the word "oracle" from the whole manuscript — including the abstract — rather than qualifying it in place.

4.3 Cost of the solve

⟨ one paragraph ⟩ Table build per orbit; sweep count to convergence; Frank–Wolfe iterations to a useful gap; memory of the vector table at the converged accumulation grid. None of this is charged to the flight budget

Table 2. Certified gap between the Frank–Wolfe lower bound and the coordinate-descent schedule, by design and budget. ⟨ fill from WP7 ⟩

Design	β	L_{FW}	J_{desc}	g_J	g_E	worst g_E	vacuous
A	1.00	⟨ value ⟩	⟨ value ⟩	⟨ pct ⟩	⟨ pct ⟩	⟨ pct ⟩	⟨ n ⟩
B	1.00	⟨ value ⟩	⟨ value ⟩	⟨ pct ⟩	⟨ pct ⟩	⟨ pct ⟩	⟨ n ⟩

— the benchmark is not flown — but it decides what is feasible to run across a campaign, and the controller of Section 7 inherits a reduced form of it, which *is* charged. [WP1, WP3, WP7]

5 Experimental setup

5.1 Field, dynamics and reference

⟨ one paragraph ⟩ GRAIL product and degree [6, 12, 16]; synthesis kernel and normalization [8, 22]; perturbation model and the conventions it follows [19, 27]; third bodies and ephemeris [1, 20]; integrator and tolerances [7, 28]; lunar orientation [18]. Adopted *reference* degree per orbit and the sense in which every result below is model-relative: what is measured is what the adopted expansion requires of its own truncations, not the distance to the true lunar field.

5.2 Populations

Table 3. Orbit populations. Designs A and B are independent scrambles of the same scrambled-Sobol coverage construction [10]; C was drawn and frozen after A and B were known to agree. That the design points are dependent by construction is why no significance level is attached to any tally. The wide-elliptic population lies outside the sampled factor box and brackets the elliptical subsatellite orbits flown on Kaguya [13].

Population	Orbits	Geometry	Role
Design A	⟨ n ⟩	⟨ box ⟩	primary
Design B	⟨ n ⟩	⟨ box ⟩	independent replication
Design C	⟨ n ⟩	⟨ box ⟩	third design
Wide-elliptic	⟨ n ⟩	perilune ⟨ km ⟩, apolune ⟨ km ⟩	regime bound
Strata (5)	⟨ n ⟩ each	inclination / argument of perilune restricted	geometry bound
Low perilune	⟨ n ⟩	⟨ km ⟩	linearization bound

5.3 Budgets

Four cost quantities appear and are never interchanged.

Table 4. Cost definitions. B_2 is the primary budget throughout; B_1 is reported only for comparability with the previous paper.

Symbol	Quantity	Note
B_1	nominal work rate, the *time average* $\langle N^2 \rangle_t = \sum_i N^2_{g(i)} \Delta t_i / T$	available without propagating; does not survive contact with the integrator. On the uniform reference output grid this is the plain mean, which is how the previous paper reported it
B_2	realized total quadratic work $\sum_{\mathrm{RHS}} N^2$	**primary**; requires propagation
B_3	measured serial gravity-kernel time	architecture-dependent; idle machine, three repeats
B_+	controller overhead, in the same units as B_2	**included in every budget reported for C-PLAN, C-LITE and C-FB**; conversion below

What B_+ is measured in. B_2 counts quadratic work, $\sum_{\mathrm{RHS}} N^2$, and B_+ is added to it, so the two must carry the same units. Two of the overhead terms already do: the pilot arc and the online probe are gravity evaluations and enter directly as $\sum N^2$. The rest — the two-body predictor, the planning sweeps, the online arg min and its bookkeeping — are processor time and are converted at the measured rate of the same run,

$$B_{+,\mathrm{eq}} \;=\; t_+ \Big/ \frac{t_{\mathrm{kernel}}}{B_2}, \tag{19}$$

with t_{kernel} the arc's measured serial gravity-kernel time. The ratio t_{kernel}/B_2 is a seconds-per-unit-work coefficient, so Eq. (19) expresses non-gravity time in N^2 units and Eq. (20) adds like to like. The coefficient is architecture-dependent and is measured under the same conditions as B_3, on an idle machine with three repeats; every table quoting B_+ names the machine it was measured on.

Normalization. β is defined at the B_1 level only, as the allocation *target* $\beta = B_1/N_{\mathrm{crit}}^2$. The realized counterpart is reported separately as $\beta_2 = B_2/(K_{\mathrm{ref}} N_{\mathrm{crit}}^2)$, with K_{ref} the comparator's right-hand-side call count on the same arc; dividing B_2 by N_{crit}^2 alone would fold the call count into the same number and is not done.

Matching convention. Comparisons are matched on total budget, and the controller's overhead comes out of its own allocation:

$$B_2 + B_+ \;=\; B_{\mathrm{tot}} \;=\; B_{2,\mathrm{F}}, \tag{20}$$

with $B_+ = 0$ for every policy that is not a controller. No fictitious overhead is added to the fixed comparator, so a controller that must plan and probe has strictly less budget left for gravity evaluation than its comparator does — the conservative reading, and the one adopted throughout.

Equation (20) is not complete until it says which side moves, and the answer is not the one the previous campaign used. There, the comparator was re-propagated to spend what the candidate had spent. That cannot be done here, because the capture fraction of Section 8 divides one difference by another and both must be taken against *the same* $E_{\text{F-OP}}$; a comparator that slides to meet each candidate would put two different F-OP instances in the numerator and the denominator.

We therefore anchor. For each orbit and each β, B_{tot} is declared in advance as the realized total quadratic work of F-OP(β) on that arc — at $\beta = 1$ that is the archived critical-degree run, so the anchor already exists and is not recomputed — and *every candidate is calibrated to it.*

That calibration costs propagations, and the cost is a consequence of the previous paper's own result rather than an inefficiency: a schedule's realized work is not known until it is flown, and calibrating on nominal per-call work misses it by a median ⟨pct⟩. Each candidate is therefore propagated, measured, its multiplier adjusted, and propagated again until $|B_2 + B_+ - B_{\mathrm{tot}}|/B_{\mathrm{tot}} \le$ ⟨pct⟩, which takes ⟨value⟩ propagations per candidate in practice. Section 8 reports the achieved match alongside every verdict.

The propagated grid is $\beta \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$; $\beta = 3$ is evaluated at benchmark level only, because at that budget the previous campaign found ⟨n⟩ of ⟨n⟩ comparisons undecidable and the experiment stops being an instrument.

5.4 Grids and their convergence

⟨one paragraph plus table⟩ The campaign runs on an accumulation grid refined on τ_{corr}, so the convergence parameter is not an absolute step but the number of samples per correlation time, $n_s = \tau_{\mathrm{corr}}(t)/\Delta t_i$, swept over $\{0.5, 1, 2, 4, 8\}$. A uniform-grid sweep at $\Delta t_{\mathrm{acc}} \in \{240, 120, 60, 30, 10\}$ s is run alongside it, both because it is the comparable quantity in the previous paper and because it isolates whether the refinement, rather than the resolution, is what matters. Three quantities are checked in each: the objective, the recovered schedule itself as a degree-profile distance, and verdict stability. The decision grid is swept separately at $\Delta t_{\mathrm{dec}} \in \{60, 120, 300\}$ s. The campaign runs at $n_s =$ ⟨value⟩. [WP4]

[direction] Section 3 predicts that the signed integral needs a finer grid than the magnitude statistic did. If the panel contradicts that — if 120 s is already converged for J and for the schedule — say so plainly and delete the prediction rather than keeping it as motivation.

5.5 Resolution rule and statistics

A comparison is decided only when the two errors differ by more than their summed reference-inclusive numerical envelope, $M_{\mathrm{res}} > 1$; comparisons below that are *undecided* and are counted for neither side. No significance level is attached to any tally: the points of a scrambled design are dependent by construction and the resolution rule censors the sample non-randomly, so a binomial test would have no valid null. Direction rests on replication across independent populations. Ratios are medians of per-orbit ratios, never ratios of medians.

5.6 Comparators

Two constant-degree comparators are carried, and they answer different questions.

- F-OP, the integer degree nearest the budget, is what a user can select in advance. Beating it is the claim.
- F-ENV, the lowest-error member of the constant family under the same budget, is found by measuring every candidate against the reference. It is a post-hoc lower envelope, not a policy; the previous paper measured it a median ⟨ value ⟩ times better than the saturating degree. It is reported to bound how much of the constant family's potential F-OP leaves unused, and no hypothesis of this paper requires beating it.

R-INT, the interior member of the radial–constant interpolation family, is carried as the strongest previously available deployable schedule.

5.7 Pre-registration and its stopping rules

Hypotheses, gates, sweeps, comparator rules, censoring and the naming rule of Section 4.2 were written to ⟨ oa01/oa02/oa03 json ⟩ and hashed before any aggregate result was inspected. Anything added later is labelled post hoc and reported separately; rejected hypotheses are reported as rejected rather than reworded. ⟨ outcome list ⟩

Two of those rules bound the campaign rather than a single number, and both are stated here because later sections invoke them.

Budget gate. The first propagated campaign is at $\beta = 1$ on one design. The budget grid is extended to the rest of the populations only if the controller also takes the resolved majority at $\beta = 0.75$ *and* $\beta = 1.25$ on a stratified sixteen-orbit subset. If it does not, the gain belongs to one budget point, the population expansion is not run, and the result is reported as such.

Screening escape. The variational screen of Section 8 is calibrated on policies that were not built by minimizing J, so its transfer to policies that were is an assumption rather than a measurement. If the propagated sign disagrees with the screened sign, the implementation is diagnosed; after two diagnostic rounds without resolution the instrument is declared uncalibrated for this class, the screen's selection is discarded, and the campaign proceeds on propagation alone with a reduced candidate list. The cost of that path is reported rather than absorbed. The rule exists so that a disagreement cannot become an open-ended loop.

6 How much is attainable, and where it comes from

This section answers RQ1 and RQ2. Nothing in it is propagated: every quantity is computed from the tabulated defect and the archived state-transition matrix, so the comparison is free of integrator noise and of the step-size effects the previous campaign had to control. What it therefore establishes is a first-order statement about a fixed reference trajectory; Section 8 tests whether the propagated arcs agree.

Predicted error, not measured error. Every error in this section is the first-order prediction $\hat{E} = \sqrt{J}$ of Eq. (6), and every ratio is written $\hat{\rho}$. The propagated quantities E and ρ appear only from Section 8 onward. The two are kept typographically distinct throughout because they answer different questions and are produced at different stages: no hypothesis stated on $\hat{E}$ is reported as though it had been tested on E. Where the resolution rule is invoked here it uses each orbit's archived numerical envelope, which is external to this paper and pre-existing, since no propagation of this campaign has yet occurred to supply one.

6.1 The two separable allocations

Two of the four rungs below are separable and are stated here once, because the Frank–Wolfe subproblem of Section 4.2 has the same structure. Both are per-interval minimizations with a single multiplier λ bisected to the budget:

$$\text{A-FORCE}: \quad \underset{N\in\mathcal{N}}{\arg\min}\Big[\textstyle\sum_{i\in I_q}\|\Delta\mathbf{a}(t_i,N)\|^2\Delta t_i+\lambda W_q N^2\Big], \qquad \text{A-SENS}: \quad \arg\min_{N\in\mathcal{N}}\Big[\textstyle\sum_{i\in I_q}\Delta t_i\,\Delta\mathbf{a}(t_i,N)^\top\mathbf{K}_i\,\Delta\mathbf{a}(t_i,N)+\lambda W \tag{21}$$

⟨ figures/fig_oa_ladder.pdf ⟩

Figure 1. The allocation ladder, one point per orbit, designs pooled. Bars are medians. Rungs 1–2 use magnitudes only; rung 3 adds the sensitivity weight; rung 4 adds the signed coupling.

⟨ figures/fig_oa_schedules.pdf ⟩

Figure 2. Degree histories at $\beta = 1$ on a representative orbit: F-OP, R-INT, A-SENS and A-SIGN. The lower panel shows the accumulated displacement each produces.

A-FORCE is the force-level benchmark of the previous paper: it never sees Φ at all. A-SENS is the natural sensitivity-weighted repair, and it is worth being precise about where its weight comes from, because two wrong answers are close at hand. Borrowing a scalar such as $\|\Phi(T,t)\mathbf{B}\|^2$ from the terminal-state proxy would be an arbitrary import, and matched to the wrong metric besides. Taking the discrete diagonal $\mathbf{u}_i^\top \mathbf{Q}_{ii}\mathbf{u}_i$ literally would make the criterion depend on the accumulation resolution, for the reason given in Section 3.3. The weight is instead the local sensitivity kernel $\mathbf{K}_i$ of Eq. (4) — the objective's own coupling evaluated on its diagonal, grid-invariant, and matched to the arc-RMS metric rather than to a terminal one.

The ladder is therefore a decomposition of one trajectory kernel and not a comparison of three unrelated criteria: A-FORCE ignores it, A-SENS retains its local diagonal $\mathbf{K}_i$, A-SIGN retains the full cross-epoch coupling. RQ2 reduces to a single question — how much of the attainable gain lives off the diagonal — and G_{sens} and G_{sign} below measure the two steps of that decomposition.

6.2 The ladder

Four allocations are compared at the same budget, each carrying one more piece of the dynamics than the last.

Table 5. Allocation ladder at $\beta = 1$. Entries are $\hat{\rho}(\text{F-OP}, \text{policy}) = \hat{E}_{\text{F-OP}}/\hat{E}_{\text{policy}}$ on the first-order prediction, medians of per-orbit ratios, so values above one favour the policy. All four are evaluated on the same reference trajectories and the same tabulated defect, and none is propagated. ⟨ fill from WP2, WP3 ⟩

	Allocation	Information used	A	B	C
1	R-INT	radius	⟨ value ⟩	⟨ value ⟩	⟨ value ⟩
2	A-FORCE	+ reference field, pointwise	⟨ value ⟩	⟨ value ⟩	⟨ value ⟩
3	A-SENS	+ diagonal kernel $\mathbf{K}_i$	⟨ value ⟩	⟨ value ⟩	⟨ value ⟩
4	A-SIGN	+ signed coupling $\mathbf{Q}$	⟨ value ⟩	⟨ value ⟩	⟨ value ⟩

RQ1 — the attainable gain. ⟨ one paragraph ⟩ A-SIGN sits a median ⟨ value ⟩ above F-OP and ⟨ value ⟩ above R-INT, the strongest deployable schedule previously available. ⟨ state whether the gain clears the numerical envelope on how ma

[direction] Gate G1 of `../ROADMAP.md` is decided here. If $\hat{\rho}(\text{R-INT}, \text{A-SIGN}) < 1.5$ at the design median, this section becomes the paper's principal result in the negative direction and Sections 7 and 8 contract sharply.

RQ2 — weight against sign. The two increments are reported separately because they are different claims. Raw error differences are not comparable across orbits of different scale, so each increment is a log ratio:

$$G_{\text{sens}} = \log \frac{\hat{E}_{\text{A-FORCE}}}{\hat{E}_{\text{A-SENS}}}, \qquad G_{\text{sign}} = \log \frac{\hat{E}_{\text{A-SENS}}}{\hat{E}_{\text{A-SIGN}}}, \tag{22}$$

the first being what correcting the weighting buys and the second what the coupling buys on top of it. ⟨ one paragraph with the two medians and their ratio. ⟩ The previous paper predicted that reweighting alone would not close the gap; ⟨ state whether that is confirmed, and by how much. ⟩

6.3 What the schedules look like

⟨ one paragraph plus figure ⟩ The signed allocation is not a monotone function of radius, which is the point: at equal budget it ⟨ describe the qualitative structure — where it spends relative to A-SENS, and whether the departure is phase-lo Reporting this matters because it is the part a deployable rule has to imitate.

6.4 Certificate

Table 2 gives the certified gap. ⟨ one paragraph ⟩ ⟨ State the median gap, the worst orbit, and therefore which of the two names
Where the gap is wide the reported ratios are conservative: the true optimum can only be better than the schedule found, so the advantage over F-OP is understated and the capture fraction of Section 7 overstated.

6.5 Sensitivity of the benchmark to its own construction

Three properties of the construction were audited and none moves a direction reported above. [WP3, WP4]

- **Start dependence.** Table 1; the spread across ⟨ n ⟩ starts is ⟨ value ⟩.
- **Accumulation grid.** Section 5.4; the schedule changes by ⟨ value ⟩ between ⟨ value ⟩ s and ⟨ value ⟩ s.
- **Degree set.** ⟨ one sentence ⟩ A coarser $\mathcal{N}$ can only make the benchmark worse, so the measured gap to the policies is conservative in that direction too.

7 A deployable controller

The benchmark of Section 6 evaluates the reference field at every epoch of a reference trajectory it already knows. This section removes that access and measures what each removal costs. This is RQ3.

What "without the reference field" means, precisely. The distinction is not between having and not having the gravity model — a propagator carries the coefficients by definition, and pretending otherwise would be artificial. It is between *global, offline model metadata* and *trajectory-specific evaluations of it*:

	Available to the controller	Withheld
Field	degree variances P_n (a table of ⟨ n ⟩ numbers)	$\Delta\mathbf{a}_{\mathrm{ref}}(t)$ along the arc
Trajectory	a cheap pilot arc it computes itself	the reference arc
Sensitivity	Φ from low-degree variational equations	the exact Φ

The controller may know the spectrum; it may not be told what the omitted acceleration is at the epochs it will fly through. That is the information boundary the rest of this section works against, and it is the one a real mission faces.

Table 6. What the benchmark uses, and what replaces it.

Benchmark uses	Replaced by	Cost, charged to B_+
Reference field, pointwise	leading omitted bands (direction) plus the measured degree-variance spectrum (amplitude)	one synthesis per look-ahead probe point; measured ⟨ pct ⟩ — the dominant term
Reference arc	cheap pilot arc, $N = 40$, loose tolerance	measured ⟨ pct ⟩ of the run's gravity work
Exact Φ	low-degree variational equations integrated on the pilot arc, $\dot{\Phi} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{G}_{40} & \mathbf{0} \end{bmatrix} \Phi$	⟨ pct ⟩

7.1 The omitted acceleration has a direction, and tabulating it is not cheap

A tail criterion predicts $\|\Delta\mathbf{a}\|$. The coupling term of Eq. (14) needs $\Delta\mathbf{a}$ itself. A controller supplied only with the magnitude can reach rung 3 of Table 5 and no further, by construction.

The direction varies on the model's own resolution scale. It is set by the leading omitted harmonics, whose half-wavelength at the spacecraft's radius is $\pi r/N$: about ⟨ value ⟩ km at $N = 120$ and ⟨ value ⟩ km at $N = 300$. A tabulation, interpolation or fitted surrogate over (r, ϕ, λ) that does not itself resolve that scale therefore cannot represent the direction, and one that does resolve it carries a number of degrees of freedom comparable to the omitted coefficients themselves, with the storage and evaluation cost that implies. This is a statement about resolution and cost, not an impossibility claim about any particular function class; Section 9 measures it directly with a surrogate built for the purpose and reports what it retains and what it costs. [WP5]

7.2 Band probe

What is affordable is to evaluate the first few omitted bands and use their sum as the direction:

$$\hat{\mathbf{v}}(t, N) \;=\; -\,\gamma(N, r) \sum_{n=N+1}^{N+k} \mathbf{a}_n(t), \tag{23}$$

with γ the ratio of the full omitted tail to the probed bands. Direction comes from the probe; amplitude from γ.

γ is taken from the model's *measured degree-variance spectrum*, not from a fitted power law:

$$\gamma(N, r) \;=\; \left[\frac{\sum_{n>N} \sigma_a^2(n; r)}{\sum_{n=N+1}^{N+k} \sigma_a^2(n; r)}\right]^{1/2}, \tag{24}$$

with $\sigma_a(n; r)$ the per-degree acceleration RMS built from the coefficient degree variances P_n. The distinction matters because the lunar spectrum is not a single power law — the previous paper measured a spectral slope of 2.13 against an effective tail exponent of 1.76 for the same field, and the gap is a property of the spectrum's shape — so a power-law extrapolation of the tail carries a systematic error that a table of P_n does not. And it costs nothing in deployability: P_n is roughly ⟨ n ⟩ numbers, a one-dimensional table.

That split is worth naming, because it is the paper's information statement in its sharpest form. **What is cheap to carry is the spectrum; what is expensive is the texture.** The amplitude of the omitted tail is a one-dimensional function of degree and can simply be uploaded. Its direction is a field on the sphere at resolution $\pi r/N$ and cannot; it has to be probed.

Cost, and why the cheap-looking figure is the wrong one. At a point where the field is *already* being evaluated to degree N, extending to $N+k$ costs $(N+k)^2 - N^2 \approx 2kN$ against N^2, that is $2k/N$ — about 5% at $N = 120$, $k = 3$ — because the associated Legendre recursion is incremental in degree and the probe inherits it.

That figure does not apply here, and saying so is the honest form of the cost accounting. A look-ahead probe is taken at a point the integrator has *not* evaluated, and the recursion for P_{nm} at a new location must run from degree zero: obtaining band n alone is not an $\mathcal{O}(n)$ operation in any standard synthesis. **Each look-ahead probe point therefore costs approximately one full synthesis**, not an increment on one.

Two properties of the construction do reduce the cost, but both act *within* a probe point rather than between them.

First, the band stack is *shared* across candidates. Evaluating the individual bands $\mathbf{a}_n$ once for n up to $N_{\max}+k$ yields $\hat{\mathbf{v}}(t, N)$ for *every* candidate $N \le N_{\max}$ by partial summation, at no further field evaluation. The candidates are therefore not independent probes but partial sums of one stack.

Second, the candidate set is a window. The offline plan supplies a nominal degree for each decision interval; writing $j(q)$ for its position in the ordered candidate grid $\mathcal{N} = \{N_1 < N_2 < \dots\}$, the online decision searches only

$$\mathcal{N}_q \;=\; \{\, N_{j(q)-\delta}, \;\dots,\; N_{j(q)+\delta} \,\}, \tag{25}$$

so that δ is a half-width in *candidate-grid indices*, not in degrees. Fixing it that way keeps the controller's complexity — $2\delta + 1$ candidates per decision — invariant when the grid is refined or coarsened; the induced span in degrees follows from the grid and is reported alongside the cost, since it is the span and not the index count that sets the band stack. The stack runs from $N_{j-\delta}+1$ to $N_{j+\delta}+k$, so the incremental cost is $\approx 2N(\Delta_{\text{span}} + k)$ with $\Delta_{\text{span}} = N_{j+\delta} - N_{j-\delta}$, rather than $2kN$.

This is the plan/probe split made concrete: the plan sets the coarse allocation, which needs the budget and the sensitivity and is smooth; the probe corrects within a window, which needs the local texture. The window is a declared parameter, swept in Section 9, and it handicaps the controller relative to the benchmark — which searches the full $\mathcal{N}$ — so the cost of the restriction appears in the capture fraction rather than being hidden.

What the overhead actually is. Covering a decision interval takes $n_{\text{probe}} \approx \Delta t_{\text{dec}}/\tau_{\text{corr}}$ probe points, and the interval contains $n_{\text{RHS}} \approx \Delta t_{\text{dec}}\, r_{\text{RHS}}$ right-hand-side calls, so the overhead is

$$\frac{n_{\text{probe}}}{n_{\text{RHS}}} \;\approx\; \frac{1}{\tau_{\text{corr}}\, r_{\text{RHS}}} \;=\; \frac{N v}{\pi r\, r_{\text{RHS}}}, \tag{26}$$

which is *independent of the decision cadence*: probing twice as often over intervals twice as short buys nothing.

⟨ figures/fig_oa_probe.pdf ⟩

Figure 3. Probe direction accuracy $\kappa = \cos\angle(\hat{\mathbf{v}}, \Delta\mathbf{a})$ against probe depth k, truncation degree N and altitude. The dashed line is the level below which the coupling term cannot be carried (Section 7.7).

Equation (26) is a local rate, and the number that belongs in a budget is its arc integral,

$$\frac{B_{\text{probe}}}{B_{\text{tot}}} = \frac{1}{N_{\text{RHS}}} \int_0^T \frac{\mathrm{d}t}{\tau_{\text{corr}}(t)} = \frac{1}{N_{\text{RHS}}} \int_0^T \frac{N(t)\, v(t)}{\pi\, r(t)}\, \mathrm{d}t, \tag{27}$$

and the two are not interchangeable. Where τ_{corr} is shortest — at perilune, where the degree is highest — the vehicle spends the least time, and the integrator's call rate is highest there as well, so both factors move together. Reading the local rate at perilune against an arc-averaged call rate overstates the cost substantially.

Evaluated on the archived reference arcs, Eq. (27) gives ⟨ pct ⟩ for a near-circular low orbit and ⟨ pct ⟩ for the eccentric geometries of Table 3, the eccentric cases being *cheaper* because their long apolune stretches need almost no probing. Those estimates hold r_{RHS} fixed at its arc average and therefore still err high; the measured values are in Table 7. The probe remains the dominant term in B_+ — against ⟨ pct ⟩ for the pilot arc and ⟨ pct ⟩ for switching — which is why it is reported as a curve rather than absorbed.

Two consequences follow, and neither is a footnote. The overhead is reported as a *measured curve* rather than a number: Table 7 gives the effective direction accuracy κ_{eff} obtained by probing at n_{probe} points, averaged over the interval, against the cost that buys it. And a cheap endpoint of that curve is declared in advance: C-LITE raises the degree of the interval's *first* right-hand-side call — a point the integrator evaluates anyway, so $2k/N$ genuinely applies — and accepts that the direction is known only at the interval's start, for τ_{corr} of its length. [WP5, WP16]

Switching is not the expensive part. A decision boundary makes the right-hand side discontinuous, and one might expect that to dominate. It does not. What the previous campaign measured is the aggregate: a radially binned schedule cost ⟨ pct ⟩ extra right-hand-side calls over seven days. Converting that to a per-switch figure requires counting the bin crossings, which we do from the archived degree histories rather than estimating; on ⟨ n ⟩ crossings it is ⟨ value ⟩ extra calls per switch. At $\Delta t_{\text{dec}} =$ ⟨ value ⟩ s there are ⟨ n ⟩ boundaries per arc, so the switching term is ⟨ pct ⟩ — roughly an order of magnitude below the probe.

Table 7. The cost–accuracy curve of the online probe, measured rather than counted. κ_{eff} is the direction accuracy averaged over the decision interval when the interval is probed at n_{probe} points; the cost column is the share of B_{tot} the probe consumes, which Eq. (20) subtracts from the controller's gravity budget. The first row is C-LITE, whose single probe is co-located with a right-hand-side call and therefore costs an increment rather than a synthesis. ⟨ fill from WP5, WP16 ⟩

Variant	n_{probe}	δ (idx)	Δ_{span} (deg)	κ_{eff}	cost, share of B_{tot}
C-LITE	1 (co-located)	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩	⟨ pct ⟩
C-PLAN	⟨ n ⟩	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩	⟨ pct ⟩
C-PLAN	⟨ n ⟩	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩	⟨ pct ⟩

⟨ one paragraph ⟩ Median κ is ⟨ value ⟩ at $k = 1$ and ⟨ value ⟩ at $k = 3$; it ⟨ degrades/holds ⟩ with altitude and ⟨ degrades/holds ⟩ with N. The refresh interval required to keep κ above ⟨ value ⟩ is ⟨ value ⟩ s, consistent with Eq. (13).

What κ does and does not measure. It is tempting to read κ as the fraction of the coupling term the probe retains. It is not, and the difference matters enough to state. The term is $2\,\mathbf{u}_i(N)^\top \mathbf{c}_i$ with $\mathbf{u}_i = \Phi\mathbf{B}\,\Delta\mathbf{a}\,\Delta t_i$, so writing $\mathbf{z} = (\Phi\mathbf{B})^\top \mathbf{c}_i$ the retained fraction is $\langle \hat{\mathbf{v}}, \mathbf{z}\rangle / \langle \Delta\mathbf{a}, \mathbf{z}\rangle$: it depends on the angles the true and estimated defects make with $\mathbf{z}$, not on the angle between them, because $\Phi\mathbf{B}$ does not preserve angles. Where $\Delta\mathbf{a} \perp \mathbf{z}$ the true contribution vanishes and any probe error produces a spurious one.

$\kappa = 1$ still forces the retained fraction to one, and a low κ still destroys it, so κ is a leading indicator rather than the quantity itself — and it has the practical advantage of being measurable without a schedule and without propagating. We therefore report both: κ as the leading indicator, and the retained fraction, computed with $\mathbf{c}_i$ from the A-SENS schedule, alongside it. *Both are diagnostics.* Neither carries a stopping rule; the deployability decision is the capture fraction of Section 8, which measures the thing itself rather than a proxy for it. [WP5]

7.3 Planning: forward–backward allocation

The coupling term $\mathbf{c}_i$ of Eq. (10) depends on the degrees chosen at *every other* epoch. A single backward sweep of an adjoint cannot supply it, because there is no schedule yet to sweep against. The controller therefore iterates.

Algorithm 2 Forward–backward degree allocation on a pilot arc

Require: pilot arc; low-degree Φ; tail model; budget B
1: $N^{(0)} \leftarrow$ separable sensitivity-weighted schedule (A-SENS form)
2: **for** $j = 1 \dots J$ **do**
3: **forward:** accumulate $\hat{\mathbf{u}}_i$ from $\hat{\mathbf{v}}(t_i, N_i^{(j-1)})$; form prefix sums
4: **backward:** form the suffix blocks $\mathbf{A}_i$ and the cross terms $\mathbf{c}_i$
5: **update:** $N^{(j)}$ by Eq. (14); bisect λ to the budget
6: **end for**
7: **return** cancellation target $\mathbf{c}$, diagonal kernel $\mathbf{K}_i$, multiplier λ

J is a free parameter and is swept in Section 9; ⟨value⟩ iterations suffice on ⟨n⟩ of ⟨n⟩ panel orbits.

7.4 Probing forward, not backward

A decision interval is $\Delta t_{\text{dec}} \approx$ ⟨value⟩ s long and the omitted direction decorrelates in $\Delta t_{\text{probe}} \approx$ ⟨value⟩ s. A rule that accumulated probes taken *inside* an interval and applied them at the *next* boundary would therefore select a degree from a direction that has already decorrelated — exactly the failure the resolution-scale argument of Section 7.1 predicts. The controller does the opposite: at each boundary it probes *ahead*, over the interval it is about to commit to.

That is legitimate, and the reason is that two different coherence questions are involved and they have different answers.

Table 8. Two coherence scales. The long-horizon failure is what forces the plan/probe split of Section 7.5; the short-horizon success is what makes forward probing admissible.

Horizon	State prediction error	Against $\pi r/N$
One decision interval, Δt_{dec}	the vehicle's own short-arc prediction, ⟨value⟩ m	far below; forward probing valid
Whole arc, pilot against flown	⟨value⟩ km after T_{coh}	far above; a pre-computed $\hat{\mathbf{v}}$ is invalid

What predicts the probe locations. At t_q the degree for the upcoming interval has not been chosen, so the integrator has not yet advanced through it, and the probe points have to come from somewhere else. Two facts make this a small requirement rather than a circularity.

First, the probe locations barely depend on the degree at all. Two candidate degrees differ in acceleration by the difference of their truncation defects, so over one interval they separate by at most $\frac{1}{2}\|\Delta\mathbf{a}\|\Delta t_{\text{dec}}^2$, which for $\|\Delta\mathbf{a}\| \sim$ ⟨value⟩ and $\Delta t_{\text{dec}} =$ ⟨value⟩ s is ⟨value⟩ mm. Against $\pi r/N$ that is ⟨value⟩ orders of magnitude below the scale on which the direction changes. The predictor therefore does not need to know N_q, which is what breaks the circle.

Second, what the predictor must achieve is a position accurate to well inside $\pi r/N$ — kilometres — not an accurate trajectory. The controller uses a two-body propagation of the current state over the interval, which requires no field evaluation and is analytic. Its error is dominated by the perturbing acceleration it omits, $\frac{1}{2}\|\mathbf{a}_{\text{pert}}\|\Delta t_{\text{dec}}^2 \approx$ ⟨value⟩ m, again far inside the tolerance. Its cost is ⟨value⟩ and enters B_+ with the rest.

`abl-predictor` replaces it with a low-degree micro-propagation at N_{plan} and reports the change in κ and in the final error, so the choice is measured rather than assumed [WP5].

The controller therefore evaluates the shared band stack at $n_{\text{probe}} = \lceil \Delta t_{\text{dec}}/\Delta t_{\text{probe}} \rceil$ points spanning the upcoming interval, along that predicted path, and sums their transported contributions to obtain each candidate's $\sum_{i \in I_q} \mathbf{u}_i(N)$. Nothing from before the boundary is reused for the direction.

Holding the degree across the interval also gives the controller exactly the switching freedom the benchmark of Section 3 has, so the two differ in information and not in resolution.

Algorithm 3 Online decision at boundary t_q

Require: plan $(\mathbf{c}, \mathbf{K}_i, \lambda, N_{\text{plan}})$; window half-width δ (in candidate-grid indices); depth k; spent work W

1: predict the state at n_{probe} points spanning $[t_q, t_{q+1})$ by two-body propagation of the current state ▷ no field evaluation; degree-independent
2: evaluate bands $N_{\text{plan}}(t_q) - \delta + 1 \ldots N_{\text{plan}}(t_q) + \delta + k$ at those points ▷ one shared stack
3: **for** $N \in \mathcal{N}_q$ **do**
4: $\quad \hat{\mathbf{v}}(t_i, N) \leftarrow -\gamma(N, r_i) \sum_{n=N+1}^{N+k} \mathbf{a}_n(t_i)$ ▷ partial sums, no new evaluation
5: **end for**
6: $N_q \leftarrow$ Eq. (14) over $\mathcal{N}_q$ with $\mathbf{c}$ from the plan
7: $\lambda \leftarrow \lambda + \text{gain} \cdot (W - W_{\text{sched}}(t_q))$ ▷ budget feedback, Section 7.6
8: hold N_q over $[t_q, t_{q+1})$

[direction] If the decision grid has to shrink toward Δt_{probe} for the coupling term to be worth anything, say so: that is a finding about the controller's required cadence, and it should be reported with the integrator cost it provokes rather than hidden by choosing Δt_{dec} after the fact. The sweep in Section 9 is what decides this.

7.5 Why the plan and the probe are split

Algorithm 2 runs on a pilot arc, which is not the arc that will be flown. Whether that matters depends on which quantity is being carried across, and the two behave differently.

The local omitted vector $\hat{\mathbf{v}}$ is field texture and decorrelates over $\pi r/N$. A pilot arc's position error exceeds that scale after $T_{\text{coh}} = \langle$ value $\rangle$, measured in Section 7.7, which is $\langle$ fraction $\rangle$ of the arc — the long-horizon row of Table 8. A schedule that fixed the probe at plan time would therefore be using a direction that is no longer the right one, and the controller does not do that: $\hat{\mathbf{v}}$ is evaluated online, at the vehicle's own state, one interval at a time (Section 7.4).

The cancellation target $\mathbf{c}$ is an accumulated quantity. The previous paper measured the defect to be coherent rather than incoherent — growth exponents of $\langle$ range $\rangle$ against the $t^{3/2}$ of a zero-mean impulse null — so the accumulated displacement is dominated by a sustained trend rather than by local texture, and it is stable between the pilot and the reference arc to $\langle$ value $\rangle$ [WP6]. The same is true of the sensitivity weight and of the multiplier.

The controller is accordingly split:

Computed offline, on the pilot arc	$\mathbf{c}$, $\mathbf{K}_i$, λ_0, N_{plan}, *indexed by orbital phase*
Computed online, ahead of each boundary	$\hat{\mathbf{v}}$ by Eq. (23), hence N_q within $\pm\delta$ of the plan

and neither half reads the reference field or the reference arc.

The plan is indexed by phase, not by time. This is forced by the same coherence argument, applied to the smooth quantities rather than to the texture. A loose $N = 40$ pilot arc drifts along track by of order $\langle$ value $\rangle$ km over seven days, which at low-lunar-orbit speed is a *timing* offset of $\langle$ value $\rangle$ s. A perilune passage lasts a few minutes. Since $\mathbf{c}$ and $\mathbf{K}_i$ are dominated by exactly those passages, a plan indexed by absolute time would apply the perilune cancellation target before or after the perilune it was computed for — an error of order the thing being corrected.

Indexing by (revolution number, phase within revolution) removes the sensitivity: the vehicle knows its own phase, and a $\langle$ value $\rangle$ s offset is $\langle$ value $\rangle$ of a revolution. It is not a return to a single-valued function of radius, because the revolution index carries the remaining horizon; that is what distinguishes the plan from the radial rules of Table 5. [WP6, WP9]

7.6 Budget feedback

A per-call budget does not survive contact with an adaptive integrator: the solver takes short steps where a concentrated schedule runs its highest degrees, so nominal equality becomes a realized overspend — a median $\langle$ pct $\rangle$ in the previous campaign, and a factor of $\langle$ value $\rangle$ at half the budget. C-FB carries the spent work as a state, accumulating N^2 over the right-hand-side calls actually made and adjusting λ so that the declared budget is met end to end rather than per call. $\langle$ one paragraph on the update law and its gain. $\rangle$ The realized-to-nominal ratio it achieves is $\langle$ value $\rangle$ against $\langle$ value $\rangle$ for the open-loop schedule.

7.7 Where the controller is not the benchmark

Four separations are measured rather than argued, and each is an ablation in Section 9: the tail amplitude in place of the true defect magnitude; the probe direction in place of the true direction; the pilot arc in place of the reference arc; and the low-degree Φ in place of the exact one. ⟨ one paragraph summarizing which dominates. ⟩

[direction] This is where a low κ would first show. It is a warning and not a stopping rule — κ is not the retained fraction, so gating on it would risk an early false negative — but if the capture fraction then confirms it, the coupling term cannot be carried, C-PLAN collapses onto rung 3, and the paper's constructive claim reduces to a sensitivity-weighted controller — which is still new relative to the previous work, but is a different and smaller claim. Draft that version of this section before the number arrives.

8 Propagated campaign

Sections 6 and 7 are first-order statements about a fixed reference trajectory. This section propagates.

8.1 Screening

Each augmented variational solve costs about one propagation, and a propagated comparison costs several, so candidates were screened before they were flown. The screening instrument is the forced variational solution, which in the previous campaign reproduced the sign of the propagated ratio on ⟨ n ⟩ of ⟨ n ⟩ resolved comparisons of a ⟨ n ⟩-orbit panel. All candidates share one reference trajectory and one gradient inside a single augmented integration and differ only through their $\Delta\mathbf{a}$, so a panel of ⟨ n ⟩ orbits is solved once per orbit with every candidate as a separate forcing channel rather than once per candidate. The selection rule, the panel and the number of candidates allowed through — two — were fixed before the screen ran. ⟨ list the candidates and the outcome. ⟩ [WP13]

One caution about the screen, stated because it is load-bearing. The previous campaign's calibration of this instrument was established on policies that were *not* constructed by minimizing J. A-SIGN is, and so are the controllers distilled from it. Scoring a candidate with the same functional it was optimized against ranks candidates fairly but does not test whether J predicts E for that class of schedule — it assumes it. That assumption is what the sign check below tests, which is why a failure there is treated as a statement about the instrument rather than as noise, and why the screening escape of Section 5.7 bounds how long it may be diagnosed.

8.2 What is propagated

Five policies are propagated at each campaign point, and two of them are there for reasons worth stating. A-SIGN is propagated because the first-order benchmark has to be tested as a trajectory statement, not assumed to be one, and because it is the denominator of the capture fraction below. R-INT is propagated because it is the strongest deployable schedule available before this paper, so beating F-OP while losing to R-INT would not be a result.

Propagated	C-PLAN, C-FB (or the two screened candidates), A-SIGN, F-OP, R-INT
Not propagated	F-ENV, constructed as a lower envelope over the constant family sweep

Each of the four calibrated policies — the two screened controllers, A-SIGN and R-INT — is brought onto the anchor B_{tot} by the iteration of Section 5.3, which is why the campaign propagates more arcs than it compares: ⟨ value ⟩ propagations per policy per orbit, ⟨ n ⟩ in total. F-OP is the anchor and is not calibrated. Table 9 reports the achieved match alongside each verdict, and no comparison is read where the match falls outside the declared band.

Arcs from the previous campaign are reused only where their realized work can be matched under the same convention; where it cannot, the arc is propagated again. Every reused arc is flagged as such in the record, because the earlier campaign matched a different budget definition and a silent reuse would charge that difference to the method. ⟨ n ⟩ arcs were reused and ⟨ n ⟩ were recomputed.

8.3 Equal realized work at $\beta = 1$

⟨ two paragraphs ⟩ The verdict against F-OP and R-INT is the claim; the row against F-ENV is reported because a reader is entitled to know how much of the constant family's potential the deployable comparator leaves unused, and it is not a policy anyone could select in advance.

Table 9. The controller against both constant-degree comparators at $\beta = 1$, matched on realized total quadratic work including the controller's own overhead B_{+}. Counts are per-orbit comparisons clearing the summed reference-inclusive envelope; ρ is the median of per-orbit ratios over all orbits of the design, not over the resolved subset. Every candidate sits on the anchor B_{tot} to within ⟨ pct ⟩; the achieved match is archived per orbit and its worst case quoted here. ⟨ fill from WP14 ⟩

	Design A			Design B		
Comparator	wins	losses	ρ	wins	losses	ρ
F-OP (deployable)	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩
R-INT (deployable)	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩
F-ENV (post hoc)	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩	⟨ n ⟩	⟨ n ⟩	⟨ value ⟩

⟨ figures/fig_oa_beta1_per_orbit.pdf ⟩

Figure 4. Orbit-by-orbit ratio at $\beta = 1$, sorted within each design. Colour records each orbit's own resolution verdict, so grey bars are comparisons the experiment cannot decide rather than ties.

⟨ figures/fig_oa_capture.pdf ⟩

Figure 5. Capture fraction across the budget grid, both designs.

8.4 Capture fraction

The controller is measured against the benchmark it approximates:

$$f \;=\; \frac{E_{\text{F-OP}} - E_{\text{C-PLAN}}}{E_{\text{F-OP}} - E_{\text{A-SIGN}}}, \qquad \text{defined only where} \quad E_{\text{A-SIGN}} < E_{\text{F-OP}} \;\; \text{and} \;\; M_{\mathrm{res}}(\text{F-OP}, \text{A-SIGN}) > 1. \tag{28}$$

Three details are fixed in advance. The benchmark A-SIGN is not deployable and carries no overhead, so it receives the full B_{tot} while C-PLAN receives $B_{\mathrm{tot}} - B_{+}$; that asymmetry is deliberate, and it is why f measures the price of deployability rather than only the quality of the approximation. Both terms are propagated errors E, never predictions, so f does not exist before this section. And $f > 1$ is possible without being an error: A-SIGN is a first-order optimum on the reference arc and can be beaten on the arc it is actually flown along, which is what Section 8.8 tests separately.

The validity condition is not cosmetic. Where the benchmark does not beat the constant degree the denominator is negative and f has no interpretation; where the two are separated by less than the numerical envelope it is a ratio of two unresolved quantities and diverges. Orbits failing either test are reported as *not applicable* and counted, never coerced to zero or to a large number. ⟨ n ⟩ of ⟨ n ⟩ orbits are eligible. ⟨ one paragraph with the median f by design and budget. ⟩

8.5 Across the budget grid

⟨ one paragraph plus table ⟩ $\beta \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$. The compute-starved end is where every deployable alternative lost in the previous campaign, so it is reported first and separately. ⟨ state the verdict at $\beta = 0.50$ explicitly, wh
[direction] H4 and H5 are decided here and are separate hypotheses on purpose: H5 ($\beta = 0.5$) can be rejected without touching H4. Report a rejection as a rejection.

8.6 Across populations

⟨ one paragraph plus table ⟩ Designs A, B, C; the five geometry strata; the wide-elliptic population, which is the regime where the previous paper's radial endpoint *won*, so a trajectory-aware controller that fails there is regime-bound; and the low-perilune subset, where the linearization behind both the benchmark and the planner is weakest and only signs are read.

8.7 Cost ladder

⟨ one paragraph ⟩ The overhead row is the one to read first: a controller that wins on error while spending its margin on its own planning has not won.

Table 10. What "equal budget" equals, down the four costs of Section 5.3, as controller over comparator. Only the primary row is matched by construction. ⟨ fill from WP16 ⟩

Cost	A	B	
Realized total quadratic work B_2	⟨ value ⟩	⟨ value ⟩	matched by construction
of which controller overhead B_+	⟨ pct ⟩	⟨ pct ⟩	pilot arc, probe, planning
Nominal per-call work B_1	⟨ value ⟩	⟨ value ⟩	reported for comparability only
Right-hand-side calls made	⟨ value ⟩	⟨ value ⟩	
End-to-end wall time	⟨ value ⟩		on ⟨ n ⟩ timed orbits
Measured kernel time B_3	⟨ value ⟩		idle machine, three repeats

8.8 Does the first-order benchmark survive propagation?

A-SIGN optimizes a linearization about a fixed reference arc. When its schedule is flown, the trajectory changes, Φ changes, and the local defect changes with them, so the schedule that was optimal for the reference arc need not be optimal for its own. The fixed-point test propagates the schedule, re-tabulates along the resulting arc, re-solves, and propagates again. ⟨ one paragraph: how far the schedule moves, how far the error moves, and after how many iterat
[WP17]

[direction] This is the test that decides whether A-SIGN is a *trajectory-level* benchmark or a *reference-arc* one. If the gain does not survive, say so in these words and rename the quantity throughout; that outcome is a strong negative result about signed allocation and is worth stating cleanly rather than burying.

8.9 Controls

Each was drawn under a rule fixed before the runs it selects, and each targets a way the result could be an artefact rather than a measurement.

Table 11. Registered controls. ⟨ fill ⟩

Control	What it tests	Outcome
Reference degree 300 → 600	whether large ratios are reference-adjacency artefacts rather than measurements	⟨ outcome ⟩
Decision-grid parity	that A-SIGN and C-PLAN switch on the same grid, so the benchmark's margin is information and not resolution	⟨ outcome ⟩
Degree-cap audit	orbits whose schedule reaches the reference degree, where its force model *is* the reference model	⟨ outcome ⟩
Phase-shifted initial states	whether the signed allocation is fitted to one arc's cancellation pattern	⟨ outcome ⟩
Time-indexed against altitude-binned schedule	whether the gain belongs to the schedule's form	⟨ outcome ⟩
Decision-grid coarsening	whether it belongs to the decision cadence	⟨ outcome ⟩
Information-source audit	that no C-PLAN run read the reference field or the reference arc	⟨ outcome ⟩

9 Ablations: which ingredient earns the gain

Each ablation removes one ingredient and leaves the rest intact, and all are scored by the forced variational solution rather than by propagation.

Two things about the panel are declared rather than left to be inferred. The fourteen ablations are run on a ⟨ n ⟩-orbit perilune-spread panel, smaller than the ⟨ n ⟩-orbit panel the candidates themselves were screened on, because each variational solve costs about one propagation and fourteen removals across the full panel would cost more than the propagated campaign it is meant to inform. And all fourteen share one augmented integration per orbit, differing only in their forcing channel, so the panel is the same for every row of Table 12

⟨figures/fig_oa_ablation_waterfall.pdf⟩

Figure 6. Ablation waterfall from the benchmark down to the constant degree.

and the rows are comparable with each other. They are *not* directly comparable with the ladder of Table 5, which stands on the full design. [WP9, WP13]

Table 12. Ablations at $\beta = 1$. "Retained" is the fraction of the controller's advantage over F-OP that survives the removal. ⟨fill⟩

Code	What is removed	Retained	
`abl-sign`	the coupling term; C-PLAN collapses to a sensitivity-weighted rule	⟨pct⟩	answer
`abl-phi`	Φ, leaving only a $(T-t)$ horizon weight (C-TGO)	⟨pct⟩	
`abl-probe`	the probe; direction replaced by a fitted (r, ϕ, λ, N) surrogate	⟨pct⟩	t
`abl-null`	the probe direction randomized at matched magnitude	⟨pct⟩	negative
`abl-kaula`	the measured spectrum P_n in γ replaced by a fitted power law	⟨pct⟩	prices the choice
`abl-pilot`	pilot arc replaced by the reference arc	⟨pct⟩	p
`abl-stm`	low-degree Φ replaced by a Keplerian one	⟨pct⟩	]
`abl-J`	$J = 1$ instead of $J = ⟨\text{value}⟩$	⟨pct⟩	ho
`abl-k`	probe depth $k = 1$ instead of $k = ⟨\text{value}⟩$	⟨pct⟩	
`abl-window`	candidate window δ narrowed and widened	⟨pct⟩	prices the
`abl-grid`	decision grid Δt_{dec} swept over ⟨value⟩ s	⟨pct⟩	the cadence the coupling
`abl-lookahead`	forward probe replaced by a probe at the boundary only	⟨pct⟩	t
`abl-predictor`	two-body probe-point predictor replaced by a low-degree micro-propagation	⟨pct⟩	pr
`abl-timeindex`	phase-indexed plan replaced by a time-indexed one	⟨pct⟩	prices the p

The negative control is the one to read first. `abl-null` keeps every magnitude and randomizes only the direction. If the controller retains its advantage under that ablation, the advantage is not coming from the signed information the paper claims it comes from, and no other row means what it appears to mean. ⟨state the outcome.⟩

The surrogate ablation tests an argument, not a variant. Section 7.1 argues that tabulating the omitted direction is not a cheap alternative, because it varies on the field's own resolution scale. `abl-probe` builds the surrogate anyway — ⟨describe: fitted on the reference field over the sampled altitude range, with ⟨n⟩ degrees of freedom⟩ — and measures what it retains. ⟨one paragraph.⟩ What is reported is the pair: what the surrogate retains, and what it costs in degrees of freedom and evaluation time to retain it. A surrogate that resolves $\pi r/N$ carries a number of parameters comparable to the omitted coefficients, ⟨value⟩ against ⟨value⟩, and that comparison is the measured form of the argument in Section 7.1 — not a claim that no such surrogate can be built.

9.1 Four controls on the design of the controller itself

The ablations above price the ingredients of C-PLAN. They cannot say whether C-PLAN is the right *shape*, because every row holds the architecture fixed. Four further controls test the architecture. None of them requires a propagation: all four run on the tabulated $\mathbf{u}_i(N)$, the suffix sequence $\mathbf{A}_i$ and the Frank–Wolfe iterate that Sections 6 and 4 produce anyway. [WP21]

Is the descent the weak link, or is the bound? The certified gap g_E conflates two quantities — how far the descent sits above the integer optimum, and how far the relaxation sits below it — and the campaign's response to a wide gap differs completely between them. On short instances ($K_{\mathrm{dec}} = ⟨\mathrm{n}⟩$, $|\mathcal{N}| = ⟨\mathrm{n}⟩$) the integer optimum J^* is obtained exactly by enumeration, which splits the gap into $J_{\mathrm{desc}} - J^*$ and $J^* - L_{\mathrm{FW}}$. ⟨state which dominates.⟩ We also report a second solver on the full problem: `S-round` rounds the Frank–Wolfe iterate — per-interval $\arg\max_N \theta_{qN}$, then sum-up rounding, then one polishing sweep — and if it beats the multi-start descent it becomes the reported schedule and the descent becomes the control.

How many directions does the coupling actually use? Each $\mathbf{M}_j$ has rank three by construction, but $\mathbf{A}_i$ is a sum of them and its eigenvalue spread has not been reported anywhere. Table ⟨tab⟩ gives the number of eigendirections carrying 95% of $\operatorname{tr} \mathbf{A}_i$ along the arc, and the objective error and schedule change when $\mathbf{A}_i$ is truncated to them. ⟨ state the number. ⟩ The consequence is architectural rather than cosmetic: if the trace concentrates in one or two directions, the accumulated state that drives the coupling is that many scalars, and a controller can carry and update them in flight instead of reading a six-vector target from an offline plan.

How far ahead does the coupling reach? Table 13 replaces $\mathbf{A}_i$ by its truncation to a horizon H and re-solves, which is A-SIGN run receding-horizon with perfect information. The curve says what a replanning controller would need to look at, and it is also the direct physical statement — absent from Section 6, where it belongs — of the range over which cancellation operates.

Table 13. Horizon sufficiency. $\hat{\rho}(H)$ is the benchmark ratio when the allocation is solved against only the next H of trajectory. H^* is the smallest H reaching 0.90 of the full-horizon ratio. ⟨ fill from WP21 ⟩

H (revolutions)	1/8	1/4	1/2	1	2	full arc
$\hat{\rho}(H)/\hat{\rho}(T)$	⟨ v ⟩	⟨ v ⟩	⟨ v ⟩	⟨ v ⟩	⟨ v ⟩	1

Could a state-feedback rule have found this without probing? This is the control that can falsify the paper's information claim, so it is stated with its failure condition first. A policy is fitted to the A-SIGN schedules on design A and evaluated on design B, and what is reported is the fraction of A-SIGN's advantage over F-OP that it retains out of sample. In the smooth-state variant the features are altitude, radius, speed, orbital phase, time to go and budget spent — position on the sphere is withheld. **If that variant retains more than 0.60 of the advantage, the gain is geometric rather than textural, the probe of Section 7 is unnecessary, and this paper's information statement is wrong.** ⟨ state the outcome, and state it as a rejection if it is one. ⟩ The second variant adds position at increasing resolution and joins the surrogate curve above. The fitted policy is a control, not a proposed method: the simplest expressive form is used and its complexity is reported alongside what it buys.

10 Discussion

10.1 What a non-separable objective changes in practice

⟨ one or two paragraphs ⟩ The separable reading of degree selection — spend where the local error is largest — is not a poor approximation to the right answer; it optimizes a different functional, and the previous paper measured that the two can order two policies oppositely. What this paper adds is that the right functional is still tractable, because the coupling has a prefix– suffix structure, and that the part a deployable rule cannot easily obtain is not the weight but the sign. The sharpest way to say it is that all three criteria are readings of one trajectory kernel: the force-level rule ignores it, the sensitivity-weighted repair retains only its local diagonal $\mathbf{K}(t,t)$, and only the third retains the cross-epoch coupling that encodes cancellation. The weight was never the missing ingredient, because the weight *is* that diagonal.

10.2 The resolution scale as a design constraint

⟨ one paragraph ⟩ One length scale, $\pi r/N$, turns out to fix five otherwise unrelated design choices:

1. the sampling requirement on the accumulation grid, so that the signed integral is not aliased (Eq. (13));
2. the resolution a surrogate for the omitted direction would have to reach, and therefore its cost (Section 7.1);
3. the probe refresh interval;
4. the probe's share of the budget, which is the arc integral of its reciprocal (Eq. (27)); and
5. the coherence horizon of the pilot arc, which is what forces the plan to be phase-indexed and the probe to be online.

The rule of thumb worth carrying away is *nothing that varies faster than the truncation's own resolution scale can be planned in advance* — and the corollary, which is the expensive half: whatever must therefore be measured in flight costs a synthesis every time the scale turns over.

10.3 Relation to goal-oriented adaptivity

⟨ one paragraph ⟩ [3]. The structural difference is that the functional here is quadratic in an integral of the residual, so the adjoint weight is not a scalar field and the resulting allocation is coupled rather than pointwise.

10.4 When would a practitioner use this?

⟨ one paragraph ⟩ The controller costs a pilot arc, an offline planning pass, and an online probe of ⟨ pct ⟩ of the gravity budget. Two of those three amortize and one does not, which decides the answer. The pilot arc and the plan are paid once per trajectory family and can be reused across a Monte Carlo ensemble or a covariance study; the probe is paid on every arc, every time. A single arc therefore pays the full setup for one use and is the worst case; ⟨ quantify the break-even in arcs. ⟩ If the campaign lands in the regime where the probe alone exceeds the margin, the honest recommendation is C-LITE or the constant degree, and Section 8 says which.

10.5 Limits

- **Model-relative.** Everything is measured against an adopted reference expansion, not against the true lunar field.
- **Linearization.** Both the benchmark and the planner are first-order. Amplitudes are not interpreted below ⟨ km ⟩ perilune, where the previous campaign found the first-order amplitude wrong by up to a factor of two while the sign survived.
- **Arc length.** Calibrated and evaluated on seven-day arcs. ⟨ state the sixty-day outcome. ⟩ A schedule planned for one arc length does not transfer to another, because the perilune descends and the dwell shifts.
- **Geometry.** ⟨ state which populations the claim holds on. ⟩
- **Certificate scope.** The gap of Section 4.2 bounds the distance to the best schedule of the linearized reference-arc problem, not of the propagated one.
- **Overhead is not small.** The online probe consumes ⟨ pct ⟩ of the budget, an order of magnitude more than the switching cost and enough to decide the comparison on its own. It is charged in full (Eq. (20)), but a reader should treat any margin narrower than it as contingent on the measured overhead rather than on the allocation idea.
- **Screening is not independent of the objective.** Candidates were screened by the same functional the benchmark optimizes (Section 8). The propagated sign check is what tests that transfer, and it is load-bearing rather than confirmatory.
- **No located optimum.** No continuous optimum in β, in k, in δ, in n_s or in J is located anywhere in this paper; all are sampled on declared grids.
- **The benchmark and the controller are not claimed with equal force.** A-SIGN is derived from the trajectory-error functional itself and is certified against it, so the attainability result stands on the derivation. C-PLAN is not claimed to be the best deployable schedule — it is the first interpretable, auditable, low-complexity one. Three of its design choices are measured rather than argued and each has a declared alternative: the coordinate descent against a global solve on short instances (Section 9), the frozen offline plan against the horizon over which the coupling actually reaches (Table 13), and the band probe against a state-feedback policy fitted to A-SIGN itself. Where a measurement favours the alternative, it is reported as favouring the alternative.

10.6 Future work

⟨ short paragraph ⟩ Each item below is named with the measurement in this paper that decides whether it is worth doing, so the list is a set of conditional branches rather than a wish list.

- *Receding-horizon replanning*, if the horizon curve of Table 13 has its knee well inside the arc or the coherence horizon of Section 7.5 is short.

- *A reduced cancellation state.* The coupling enters only through $\mathbf{A}_i$, whose rank is at most three by construction and whose eigenvalue spread is reported in Section 9. If the trace concentrates in one or two directions, the accumulated state the controller must carry is that many scalars rather than a six-vector, and it can be updated in flight instead of being planned offline — a cheaper controller than the one flown here, and the same reduction that makes a trellis formulation tractable.
- *A switching penalty*, which turns the coordinate descent into a trellis problem and is only worth the state if that reduction holds.
- *A vector-tail surrogate* in place of the probe, along the cost–retention curve of Section 9, if the measured probe overhead stays at the level reported here.
- *A stronger solver* — relax-and-round on the Frank–Wolfe iterate, or a global solve — if the gap decomposition attributes the certified gap to the descent rather than to the relaxation.
- Allocation against an ensemble of initial states rather than one; transfer to other fields [15]; and propagating coefficient uncertainty into the degree-variance spectrum the amplitude estimate rests on.

11 Conclusion

[direction] Three versions of this section are drafted below, one per outcome band of `../OUTCOMES.md`. Keep exactly one before submission and delete the other two together with this note. Drafting them in advance is deliberate: the point of the pre-registration is that the wording does not get chosen after the numbers arrive.

Allocating a spherical-harmonic degree budget from the local force error optimizes the wrong functional. Written in the form the dynamics impose, the objective takes the norm after a signed integral against the state-transition matrix, which couples every pair of epochs and makes the allocation non-separable. The classical multiplier treatment and its sensitivity-weighted repair both still *work* — they are separable and a single multiplier decouples them — but they optimize the wrong functional, and the discretized form says exactly how: all three criteria are readings of one trajectory kernel, which the force-level rule ignores, the sensitivity-weighted repair reduces to its local diagonal, and only the third keeps whole. The weight was never the missing ingredient, because the weight *is* that diagonal.

The coupling nonetheless has a prefix–suffix structure that keeps a full sweep over the decision intervals at $\mathcal{O}(M|\mathcal{N}|)$ and admits a Frank–Wolfe bound on its convex-hull relaxation, so the problem is solved and certified without propagating anything, at a certified error gap of ⟨pct⟩. On ⟨n⟩ orbits of ⟨n⟩ independent designs the certified allocation is a median ⟨value⟩ better than the equal-budget constant degree and ⟨value⟩ better than the strongest previously available deployable schedule, and ⟨pct⟩ of that advantage lives off the diagonal.

Carrying it into a deployable rule turns on a quantity a tail criterion does not provide: the *direction* of the omitted acceleration. That direction varies on the field's own resolution scale $\pi r/N$, which is ⟨value⟩ km at degree 300, so a surrogate coarser than that scale cannot represent it and one fine enough to do so carries a number of degrees of freedom comparable to the omitted coefficients themselves. It can be probed, and the probe is the controller's dominant cost rather than a rounding term: covering a decision interval takes a measured ⟨pct⟩ of the total budget, and recovers the direction to $\kappa =$ ⟨value⟩. The same length scale sets how far ahead the probe may look and how long a plan made on a cheap pilot arc stays valid, and those two answers differ: forward over one decision interval, yes; across a seven-day arc, no. That is why the controller plans its cancellation target offline and probes forward one interval at a time rather than uploading a schedule.

Held to equal realized quadratic work including its own overhead, that controller has the smaller seven-day position error than the constant degree on ⟨n⟩ of ⟨n⟩ resolved comparisons across ⟨n⟩ populations, capturing a median ⟨value⟩ of the certified benchmark's advantage. ⟨state the $\beta = 0.5$ outcome explicitly.⟩ ⟨state the wide-elliptic outcome explicitly.⟩

Every conclusion is model-relative and bounded by the tested designs, arc lengths and budgets. ⟨closing sentence naming th

Acknowledgments

⟨to be written⟩

Data availability

⟨ archive DOI, manifest digest chain, pre-registration hashes ⟩

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